

Tian Bai

Statistics Ph.D Student at Stanford University Stanford, CA
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EDUCATION

Stanford University, Stanford, CA, United States 09/2025 – Present
Ph.D. in Statistics Cumulative GPA: 4.05

McGill University, Montreal, QC, Canada 09/2021 – 12/2024
Bachelor of Science in Honours Mathematics and Computer Science Cumulative GPA: 4.00
Minor in Statistics Graduated with First-Class Honours

RESEARCH

My research interests lie broadly in statistics and machine learning, with an emphasis on developing reliable, interpretable, and trustworthy AI/ML systems. I am particularly interested in uncertainty quantification, statistical inference with black-box models, and principled approaches to understanding complex learning systems. My current work centers on conformal inference, while I am also exploring topics including causal inference, empirical Bayes methods, and multiple testing.

8. Bai T*, Almeida A*, Lei L. Stein’s Unbiased Risk Estimate for Heteroskedastic Populations. *In Preparation*.
Summary: Develops a generalized SURE formula for heteroskedastic populations, enabling direct risk minimization when selecting hyperparameters such as shrinkage coefficients and KDE bandwidths.
7. Bai T, Rothenhäusler D. Transporting Causal Estimands across Populations with Minimax Optimality. *In Preparation*.
Summary: How can we transport causal conclusions from one study population to another? We develop minimax-optimal methods leveraging auxiliary response information from the target population.
6. Bai T, Jin Y. Conformal Selective Prediction with General Risk Control. *Under review at the Journal of the American Statistical Association, 2026*.
Summary: Introduces a general conformal framework for selective prediction that provides finite-sample, distribution-free control for arbitrary risk measures via multiple testing with generalized e-values.
5. Shi W, Li Y, Bai T, Zhang Y, Zhao K, Legault MA, Peng H, Zhao Y, Kolaczyk ED, Yu X, Yang A. Beyond Go/No-Go Decisions: A Regional Selection Framework for Uncertainty-Aware Molecule Screening. *ChemRxiv preprint, 2026*.
Summary: Develops a flexible three-way, uncertainty-aware framework for compound screening that moves beyond binary go/no-go decisions.
4. Bai T, Tang P, Xu Y, Svetnik V, Khalili A, Yu X, Yang A. Conformal Selection for Efficient and Accurate Compound Screening in Drug Discovery. *Journal of Chemical Information and Modeling, 2025*.
Summary: Applies conformal selection to compound screening, delivering distribution-free error control together with improved screening efficiency and accuracy.
3. Bai T, Zhao Y, Yu X, Yang A. Multivariate Conformal Selection. *Forty-Second International Conference on Machine Learning (ICML 2025)*.
Summary: Extends conformal selection to multiple response variables, providing finite-sample risk control under complex, multidimensional selection criteria.
2. Bai T, Jin Y. Optimized Conformal Selection: Powerful Selective Inference After Conformity Score Optimization. *Under review (minor revision) at the Journal of the Royal Statistical Society: Series B, 2024*.
Summary: Enables data-driven conformity-score optimization and model selection prior to candidate selection while preserving distribution-free FDR control and improving statistical power.
1. Grant L, Diagne M, Aroutiunian R, Hopkins D, Bai T, Kondrup F, Clark G. Machine learning outperforms the Canadian Triage and Acuity Scale (CTAS) in predicting need for early critical care. *Canadian Journal of Emergency Medicine, 2024*.
Summary: Uses interpretable machine learning to predict early critical-care needs in emergency departments, outperforming the Canadian Triage and Acuity Scale.

CONTRIBUTED TALKS

- Conformal Selective Prediction with General Risk Control
 - ICSA Applied Statistics Symposium, Student Paper Award Session 06/2026
 - Berkeley-Stanford Joint Colloquium, 2026 04/2026
- Model-Free Selection Inference for Drug Discovery via Conformal Prediction
 - McGill Seventh Undergraduate Research Conference 08/2024

WORK EXPERIENCE

Vffice, Inc.

Software Developer Intern

Longueuil, QC, Canada

05/2023 – 08/2023

Developed and integrated custom business management solutions using Microsoft Azure and C#.

Mentor: Yisheng Cao

AWARDS AND HONOURS

William R. and Sara Hart Kimball Fellow	Stanford University	2025
ICSA Student Paper Award	ICSA	2026
ICSA Travel Award	ICSA	2026
First-Class Honours	McGill University	2025
Charles Fox Memorial Prize	McGill University	2025
Science Undergraduate Research Award (SURA)	McGill University	2024
Sir Edward Beatty Memorial Scholarship in Mathematics	McGill University	2024
Tomlinson Engagement Award for Mentoring	McGill University	2023, 2024
A. D. Pelletier Scholarship in Mathematics and Statistics	McGill University	2023
Robert W. Wilson Scholarship	McGill University	2022
Dean's Honour List	McGill University	2022, 2023, 2025

TEACHING

Stanford University

- *Teaching Assistant*, DATASCI 112: Principles of Data Science Winter 2026

McGill University

- *Teaching Assistant*, MATH 350: Honours Discrete Mathematics Fall 2023
- *Teaching Assistant*, MATH 457: Honours Algebra 4 Winter 2024
- *Mathematics Helpdesk Tutor* Fall 2023
- *Computer Science Helpdesk Tutor* Winter 2024

SERVICE

- Journal Reviewing: TPAMI, Statistics and Computing
- Conference Reviewing: NeurIPS 2026

EXTRACURRICULAR ACTIVITIES

- Competitive Programming: Contestant with [Compete McGill](#). Team ranked 5th in the 2023 NAQ.

SKILLS

- Programming: Python, R, C/C++, SQL
- Software/Tool: L^AT_EX, Git, Powershell, Rstudio